

DocChat Test Document

What is DocChat?

DocChat is a retrieval-augmented generation (RAG) system that answers questions using the content of uploaded PDF documents. It combines a vector database for semantic search with a large language model for generating grounded, cited answers.

How Retrieval Works

When a document is uploaded, its text is extracted and split into small overlapping chunks. Each chunk is converted into a vector embedding that captures its meaning. These embeddings are stored in a vector database called ChromaDB. When a user asks a question, the question is embedded the same way, and the chunks with the closest matching vectors are retrieved.

The Agent and Tool Use

The retrieved chunks are passed to a language model along with the user's question. The model is also given access to a web search tool. If the retrieved chunks do not contain enough information to answer the question, the model can choose to call the web search tool to find current information instead of guessing.

Why Chunking Matters

Chunk size affects retrieval quality significantly. If chunks are too large, retrieval becomes vague and wastes tokens. If chunks are too small, important context can be lost. Overlapping chunks help prevent an idea from being split awkwardly across a chunk boundary.

Project Goal

This project was built as a hands-on way to learn core AI engineering concepts: document ingestion, chunking strategy, embeddings, vector search, and agentic tool use with a language model.